

# FPRO7-A1190-OSB

Serie FPRO · R32 · senza serbatoio / without tank



FPRO7-A1190-OSB

RESA FRIGORIFERA · COOLING CAPACITY

**785** kW

EER · 12/7 °C · AIR 35 °C

**2,92**

RESA NOMINALE · 15 °C · AIR 25 °C

**1.184** kW

PORTATA ACQUA · WATER FLOW

**135,0** m<sup>3</sup>/h

|                            |               |                                        |                       |
|----------------------------|---------------|----------------------------------------|-----------------------|
| Compressori / Compressors  | 6 × scroll    | L × P × H (mm)                         | 8.120 × 2.197 × 2.442 |
| Circuiti / Circuits        | 2             | Serbatoio / Tank                       | —                     |
| Ventilatori / Fans         | 14            | Taglia / Size                          | 1190                  |
| Free cooling (ΔT 8 K)      | 960 kW *      | Campo aria esterna / Ambient air range | -5 / +47 °C           |
| Refrigerante / Refrigerant | R32 · GWP 675 |                                        |                       |

## Resa in sintesi / Capacity at a glance <sup>kW</sup>

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 897   | 843   | 785   | 727   | 664   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 1.184 | 1.112 | 1.039 | 962   | 886   |

Refrigeratori condensati ad aria con free cooling integrato: quando l'aria esterna è abbastanza fredda il sistema raffredda l'acqua da solo, con i compressori fermi.

Air-cooled chillers with integrated free cooling: when the outdoor air is cold enough the system cools the water on its own, with the compressors off.

# Dati tecnici

## Technical data

FPR07-A1190-0SB

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.184,10 kW              |
| Resa frigorifera / Cooling capacity | 1.018.326 kcal/h         |
| Potenza assorbita / Power input     | 211,91 kW                |
| EER / EER                           | 5,59                     |
| Portata acqua / Water flow          | 203,66 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 785,01 kW                |
| Potenza assorbita / Power input     | 268,45 kW                |
| EER / EER                           | 2,92                     |
| ESEER / ESEER                       | 4,16                     |
| Portata acqua / Water flow          | 135,02 m <sup>3</sup> /h |

### Componenti / Components

|                                               |                     |
|-----------------------------------------------|---------------------|
| Compressori / Compressors                     | 6 × Scroll          |
| Circuiti frigoriferi / Refrigerant circuits   | 2                   |
| Ventilatori / Fans                            | 14                  |
| Free cooling (ΔT 8 K) / Free cooling (8 K ΔT) | 960 kW *            |
| Ventilatori free cooler / Free cooler fans    | 16                  |
| Condensatore / Condenser                      | a microcanali       |
| Evaporatore / Evaporator                      | a fascio tubiero    |
| Refrigerante / Refrigerant                    | R32 • GWP 675 • A2L |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 8.120 × 2.197 × 2.442 mm  |
| Campo aria esterna / Ambient air range                                                    | -5 / +47 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

\* Potenza di free cooling indicativa, da confermare in fase di offerta. / Indicative free cooling capacity, to be confirmed at quotation stage.

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 • 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 769,3 kW     |
| MEG 30 % | × 0,950 | 745,8 kW     |
| MEG 40 % | × 0,920 | 722,2 kW     |
| MPG 20 % | × 0,975 | 765,4 kW     |
| MPG 30 % | × 0,960 | 753,6 kW     |
| MPG 40 % | × 0,950 | 745,8 kW     |

# Prestazioni Performance

FPRO7-A1190-OSB · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  · 12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15    | 20    | 25    | 30    | 35    | 40  | 45  |
|---------------|-------|-------|-------|-------|-------|-----|-----|
| 1             | 808   | 766   | 723   | 676   | 628   | 580 | 530 |
| 2             | 838   | 795   | 750   | 702   | 653   | 602 | 551 |
| 3             | 868   | 824   | 777   | 727   | 678   | 624 | 573 |
| 4             | 901   | 855   | 807   | 755   | 704   | 649 | 596 |
| 5             | 934   | 886   | 836   | 783   | 730   | 674 | 618 |
| 6             | 968   | 918   | 867   | 813   | 758   | 700 | 641 |
| 7             | 1.002 | 951   | 897   | 843   | 785   | 727 | 664 |
| 8             | 1.039 | 985   | 931   | 874   | 815   | 753 | 690 |
| 9             | 1.075 | 1.020 | 964   | 906   | 844   | 779 | 716 |
| 10            | 1.113 | 1.057 | 999   | 938   | 874   | 809 | 743 |
| 11            | 1.150 | 1.094 | 1.034 | 970   | 905   | 839 | 771 |
| 12            | 1.190 | 1.132 | 1.070 | 1.005 | 937   | 868 | 798 |
| 13            | 1.231 | 1.170 | 1.106 | 1.040 | 969   | 898 | 825 |
| 14            | 1.273 | 1.210 | 1.145 | 1.076 | 1.004 | 930 | 855 |
| 15            | 1.315 | 1.250 | 1.184 | 1.112 | 1.039 | 962 | 886 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  |
|---------------|-----|-----|-----|-----|-----|-----|-----|
| 1             | 176 | 196 | 217 | 241 | 267 | 297 | 331 |
| 2             | 176 | 196 | 217 | 241 | 267 | 297 | 332 |
| 3             | 175 | 196 | 217 | 241 | 268 | 298 | 332 |
| 4             | 175 | 195 | 217 | 241 | 268 | 298 | 332 |
| 5             | 174 | 195 | 217 | 241 | 268 | 298 | 332 |
| 6             | 173 | 195 | 217 | 242 | 268 | 298 | 332 |
| 7             | 173 | 194 | 217 | 242 | 268 | 299 | 332 |
| 8             | 172 | 194 | 217 | 242 | 269 | 299 | 333 |
| 9             | 170 | 193 | 216 | 241 | 269 | 299 | 333 |
| 10            | 169 | 192 | 216 | 241 | 269 | 299 | 333 |
| 11            | 168 | 191 | 215 | 241 | 269 | 299 | 333 |
| 12            | 166 | 190 | 215 | 241 | 268 | 299 | 333 |
| 13            | 165 | 189 | 214 | 240 | 268 | 299 | 333 |
| 14            | 163 | 188 | 213 | 239 | 268 | 299 | 333 |
| 15            | 161 | 186 | 212 | 239 | 268 | 299 | 333 |

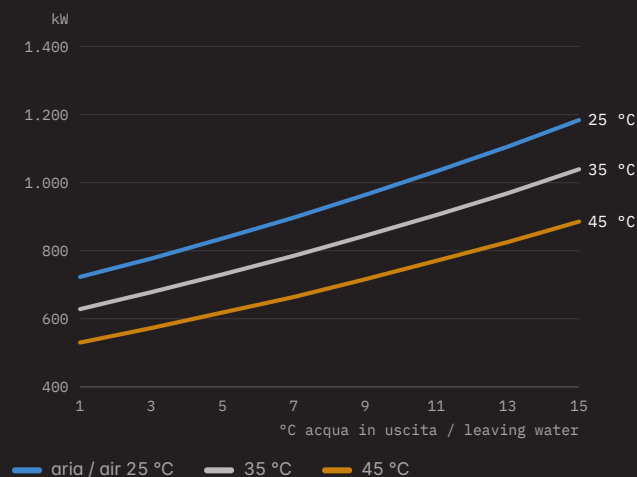
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   |
|---------------|------|------|------|------|------|------|------|
| 1             | 4,59 | 3,91 | 3,33 | 2,81 | 2,35 | 1,95 | 1,60 |
| 2             | 4,77 | 4,06 | 3,45 | 2,91 | 2,44 | 2,03 | 1,66 |
| 3             | 4,95 | 4,21 | 3,57 | 3,02 | 2,53 | 2,10 | 1,73 |
| 4             | 5,16 | 4,38 | 3,71 | 3,13 | 2,63 | 2,18 | 1,79 |
| 5             | 5,36 | 4,54 | 3,85 | 3,24 | 2,73 | 2,26 | 1,86 |
| 6             | 5,58 | 4,72 | 3,99 | 3,37 | 2,82 | 2,35 | 1,93 |
| 7             | 5,81 | 4,89 | 4,14 | 3,49 | 2,92 | 2,43 | 2,00 |
| 8             | 6,06 | 5,09 | 4,29 | 3,62 | 3,03 | 2,52 | 2,07 |
| 9             | 6,31 | 5,28 | 4,45 | 3,75 | 3,14 | 2,60 | 2,15 |
| 10            | 6,58 | 5,50 | 4,63 | 3,89 | 3,25 | 2,70 | 2,23 |
| 11            | 6,85 | 5,72 | 4,80 | 4,02 | 3,37 | 2,80 | 2,31 |
| 12            | 7,16 | 5,95 | 4,98 | 4,18 | 3,49 | 2,90 | 2,39 |
| 13            | 7,48 | 6,19 | 5,17 | 4,33 | 3,61 | 3,00 | 2,48 |
| 14            | 7,83 | 6,45 | 5,38 | 4,49 | 3,75 | 3,11 | 2,57 |
| 15            | 8,19 | 6,72 | 5,59 | 4,66 | 3,88 | 3,22 | 2,66 |

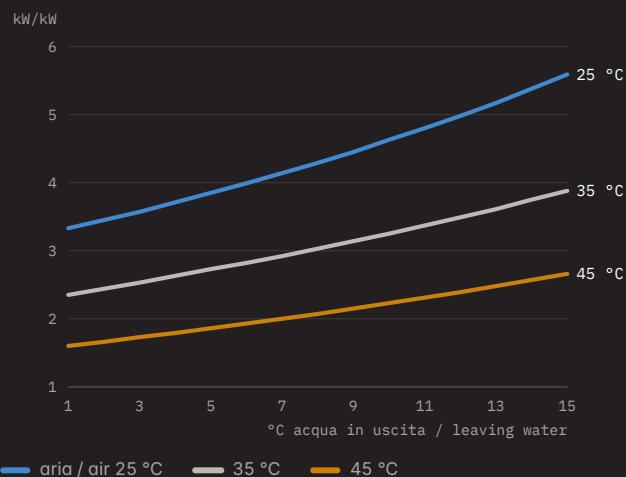
# Curve e dimensioni

## Curves and dimensions

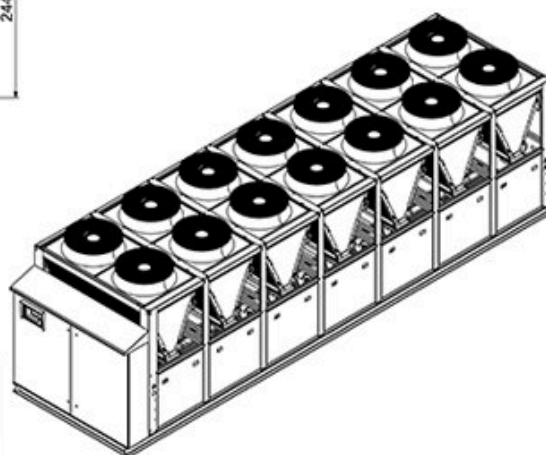
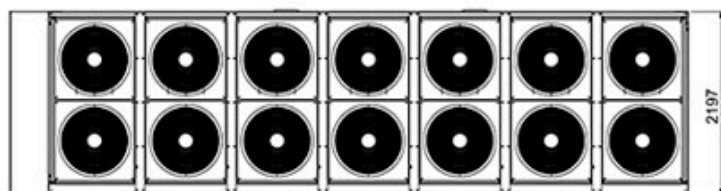
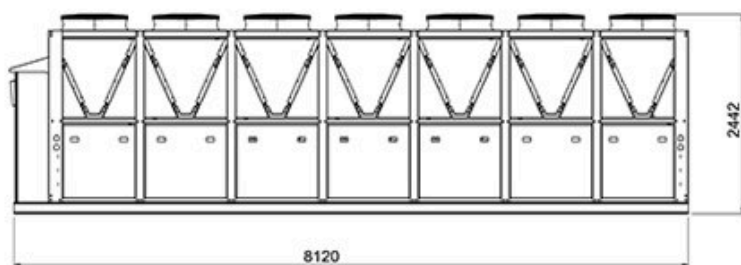
### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 8.120 × 2.197 × 2.442 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

# Optional — 1/2 Options

FPRO7-A1190-OSB · 65 optional disponibili per questa configurazione · 65 options available for this configuration

## Pompe / Pumps

Pompa 132 m³/h – prevalenza 25 m

Pump 132 m³/h – head 25 m

Pompa 132 m³/h – prevalenza 40 m

Pump 132 m³/h – head 40 m

Pompa doppia montata 132 m³/h – prevalenza 25 m

Twin pump, factory-fitted, 132 m³/h – head 25 m

Pompa doppia montata 132 m³/h – prevalenza 40 m

Twin pump, factory-fitted, 132 m³/h – head 40 m

## Serbatoi e vasi di espansione / Tanks and expansion vessels

Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia in AISI 304 montata

External water tank 2000 l, AISI 304 with factory-fitted twin AISI 304 pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata

External water tank 3000 l, AISI 304 with factory-fitted twin AISI 304 pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata

External water tank 5000 l, AISI 304 with factory-fitted twin AISI 304 pump

Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia montata

External water tank 2000 l, AISI 304 with factory-fitted twin pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata

External water tank 3000 l, AISI 304 with factory-fitted twin pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata

External water tank 5000 l, AISI 304 with factory-fitted twin pump

Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 1500 l, AISI 304 with factory-fitted single AISI 304 pump

Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 2000 l, AISI 304 with factory-fitted single AISI 304 pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 3000 l, AISI 304 with factory-fitted single AISI 304 pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 5000 l, AISI 304 with factory-fitted single AISI 304 pump

Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola montata

External water tank 1500 l, AISI 304 with factory-fitted single pump

Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola montata

External water tank 2000 l, AISI 304 with factory-fitted single pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata

External water tank 3000 l, AISI 304 with factory-fitted single pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata

External water tank 5000 l, AISI 304 with factory-fitted single pump

Serbatoio idraulico esterno 2000 l pressurizzato in acciaio zincato con pompa doppia montata

External water tank 2000 l, pressurised, galvanised steel with factory-fitted twin pump

Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa doppia montata

External water tank 3000 l, pressurised, galvanised steel with factory-fitted twin pump

Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa doppia montata

External water tank 5000 l, pressurised, galvanised steel with factory-fitted twin pump

Serbatoio idraulico esterno 1500 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 1500 l, pressurised, galvanised steel with factory-fitted single pump

Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 3000 l, pressurised, galvanised steel with factory-fitted single pump

Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 5000 l, pressurised, galvanised steel with factory-fitted single pump

Vaso di espansione 100 l

Expansion vessel 100 l

Vaso di espansione 200 l

Expansion vessel 200 l

Vaso di espansione 300 l

Expansion vessel 300 l

Vaso di espansione 500 l

Expansion vessel 500 l

Vaso di espansione 750 l

Expansion vessel 750 l

## Valvole e filtri / Valves and filters

Elettrovalvola 2"

Solenoid valve 2"

Elettrovalvola 3"

Solenoid valve 3"

Elettrovalvola 4"

Solenoid valve 4"

Elettrovalvola 5"

Solenoid valve 5"

Elettrovalvola 6"

Solenoid valve 6"

Filtro a dischi in acciaio 2"

Steel disc filter 2"

# Optional — 2/2

## Options

FPRO7-A1190-OSB · 65 optional disponibili per questa configurazione · 65 options available for this configuration

### Valvole e filtri / Valves and filters

Filtro a dischi in acciaio 3"

Steel disc filter 3"

Filtro a dischi in acciaio 4"

Steel disc filter 4"

Filtro meccanico a Y 2"

Y-strainer 2"

Filtro meccanico a Y 3"

Y-strainer 3"

Filtro meccanico a Y 4"

Y-strainer 4"

Filtro meccanico a Y 5"

Y-strainer 5"

Filtro meccanico a Y 6"

Y-strainer 6"

Valvola a 3 vie 2" miscelatrice

3-way mixing valve 2"

Valvola a 3 vie 3" miscelatrice

3-way mixing valve 3"

Valvola a 3 vie 4" miscelatrice

3-way mixing valve 4"

Valvola a 3 vie 5" miscelatrice

3-way mixing valve 5"

Valvola a 3 vie 6" miscelatrice

3-way mixing valve 6"

Valvola a 3 vie 2" on/off

3-way valve 2" on/off

Valvola a 3 vie 3" on/off

3-way valve 3" on/off

Valvola a 3 vie 4" on/off

3-way valve 4" on/off

Valvola a 3 vie 5" on/off

3-way valve 5" on/off

Valvola a 3 vie 6" on/off

3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori

Compressor acoustic enclosure

Compressore inverter

Inverter compressor

Desurriscaldatore a piastre

Plate desuperheater

Rilevatore fughe gas

Refrigerant leak detector

### Ventilazione e clima / Fans and climate

Inverter ventilatori

Fan inverter

Ventilatori EC

EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)

Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)

RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)

RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete

Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza

evaporatore + termostato) fino a -25 °C

Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla

Spring anti-vibration mounts

Antivibranti in gomma

Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
  - Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
  - Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
  - Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
  - Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
  - Free cooling: potenza riferita a una differenza di 8 K tra acqua e aria esterna; valore indicativo, da confermare in fase di offerta.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
  - With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
  - Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
  - Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
  - Procryon reserves the right to change data and features without notice, as part of continuous product improvement.
  - Free cooling: capacity referred to an 8 K difference between water and outdoor air; indicative value, to be confirmed at quotation stage.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

Procryon — Smart Cooling Engineering S.r.l.  
Via Ugo Vaglia 10 — Gavardo (BS) — Italia  
[www.procryon.com](http://www.procryon.com)



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Continuity by design